

Multilevel Inheritance

Basic Information

Title: Multilevel Inheritance

File Name: Multilevel_inheritance.java

Objective / Aim

To demonstrate multilevel inheritance in Java, where inheritance occurs through multiple levels.

System Requirements

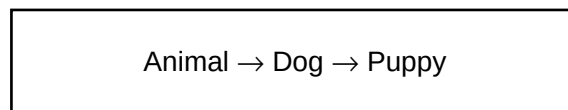
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create Animal as the base class.
3. Create Dog extending Animal.
4. Create Puppy extending Dog.
5. Create an object of Puppy.
6. Call methods from all inheritance levels.
7. End the program.

UML Class Diagram



Source Code

```
// Base class
class Animal {

    void eat() {
        System.out.println("Animal is eating");
    }
}

// Dog inherits Animal
class Dog extends Animal {

    void bark() {
        System.out.println("Dog is barking");
    }
}

// Puppy inherits Dog
class Puppy extends Dog {

    void play() {
        System.out.println("Puppy is playing");
    }
}

public class Multilevel_inheritance{
    public static void main(String[] args) {

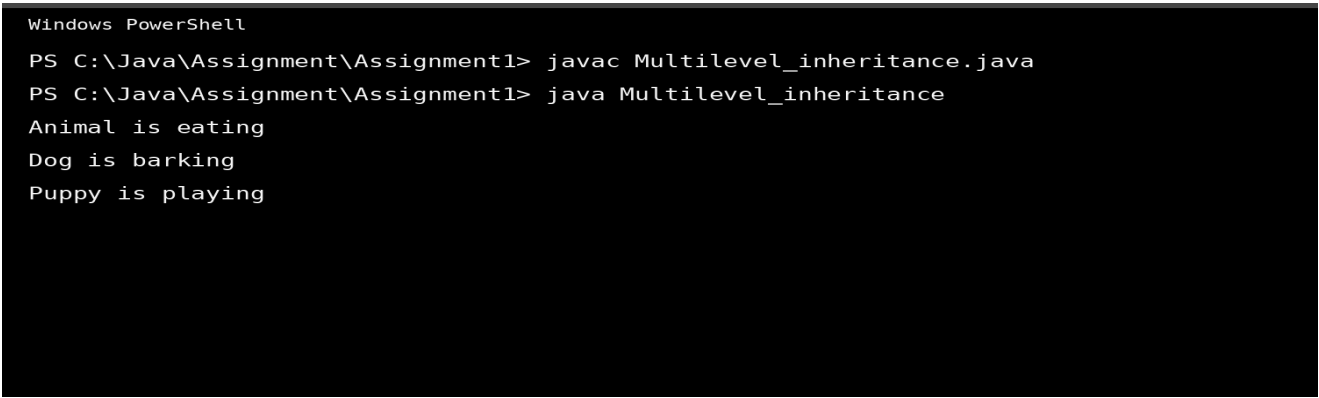
        // Puppy can access methods from all levels
        Puppy puppy = new Puppy();

        puppy.eat();
        puppy.bark();
        puppy.play();
    }
}
```

Compilation & Execution Commands

```
javac Multilevel_inheritance.java
java Multilevel_inheritance
```

Output Screen



```
Windows PowerShell
PS C:\Java\Assignment\Assignment1> javac Multilevel_inheritance.java
PS C:\Java\Assignment\Assignment1> java Multilevel_inheritance
Animal is eating
Dog is barking
Puppy is playing
```

Result

Thus, the Multilevel Inheritance program was implemented successfully using Java inheritance concepts.